Kadhir Ponnambalam

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EDUCATION

Bachelor of Engineering Co-op (BEng) | Engineering 1| 2025 - 2029 **GPA: 12.0**
McMaster University - Entrance Scholarship $3500

WORK EXPERIENCE

Research Assistant **Oct 2025 – Present**
Computer Engineering Lab | McMaster: Dr. Abdelhadi

- Integrated and evaluated open-source data processing software (CaImAn) on constrained Linux systems, benchmarking performance, memory usage, and throughput for deployment feasibility
- Analyzed system bottlenecks and documented trade-offs between compute, latency, and accuracy for deployment
- Produced technical documentation for system architecture, experimental results, and optimization recommendations

Research Assistant **Jun – Sept 2023**
Electrical and Computer Engineering Lab | McMaster: Dr. Jamal Deen

- Conducted experimental and sensor-based engineering work using microcontrollers and electronic components, collaborating with graduate researchers and contributed to peer-reviewed publications

PUBLICATIONS

1) Taheri, M., **Ponnambalam**, **K**., Shboul, A. A., Ketabi, M., & Deen, M. J. (2025). Novel Nickel Oxide/Graphene Composite Sensor: A Low-Temperature Approach to Flexible Temperature Sensing. *Advanced Materials Technologies*, e00060.

2) Taheri, M., **Ponnambalam**, **K**., Al Shboul, A. M., Ketabi, M., & Deen, M. J. (2025). Nickel oxide/graphene-based temperature sensor: Flexible design with low processing temperatures and high stability. *Sensors and Actuators A: Physical*, 116693.

PROJECTS

Autonomous Anomaly Detection for Logging-system **Jan 2026**
[github.com/kadgitub7/Anomaly-detection-for-Logging-system][youtube.com/watch?v=TJmfjUQGjUo&t=152s]

- Designed and implemented a distributed backend system using Flask REST APIs to ingest telemetry-style log data from a web client
- Built a scalable data pipeline using HDFS, Spark, and SQL to process, store, and serve structured analytic data
- Implemented logging, error handling, and API endpoints, to expose system metrics and processed outputs for downstream monitoring

EZ-MLAI (Automated machine learning models with performance and live prediction capabilities) **Jun 2024**
[github.com/kadgitub7/ez-mlai-Automated-Machine-Learning-] [youtube.com/watch?v=PeN1xNjkYBw]

- Developed a backend-driven web platform with REST APIs enabling dataset upload, model execution, and persistent access to results
- Implemented backend workflows for data validation, job execution, result storage, and live prediction

SKILLS

Programming: Python, SQL, Bash, Java
Backend: Flask, REST API design, data validation, logging, error handling
Systems: Linux, distributed data pipelines, telemetry-style data modeling
Cloud & DevOps: Docker, Git
Data: SQL databases, structured and unstructured data